# Power BI Project

## I. Project Overview, Business Scenario, and Dataset Design

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## 1. Project Title

**Retail Sales Performance Dashboard**

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## 2. Business Scenario

You are working as a data analyst for a retail company that sells products across multiple regions and categories.

The management wants a dashboard to answer:

- How much are we selling?
- Which products perform best?
- Which regions generate the most revenue?
- How are sales trending over time?
- Who are our top customers?

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## 3. Business Objectives

By the end of the project, the dashboard should provide:

1. Total Sales, Profit, and Quantity
2. Sales by Category and Subcategory
3. Sales by Region and City
4. Monthly and Yearly Trends
5. Top 10 Products and Customers
6. Basic KPI indicators

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# 4. Dataset Design

We will use a **star schema model** (important concept for beginners).

### Tables:

1. Sales (Fact Table)
2. Customers (Dimension)
3. Products (Dimension)
4. Date (Calendar Table)
5. Regions (Dimension)

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# 5. Full Dataset Structure

## 5.1 Sales Table (Main Fact Table)

Columns:

- OrderID
- OrderDate
- CustomerID
- ProductID
- RegionID
- Quantity
- UnitPrice
- Discount (0 to 0.3)
- SalesAmount
- Profit

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## 5.2 Customers Table

- CustomerID
- CustomerName
- Segment (Consumer, Corporate, Home Office)

---

## 5.3 Products Table

- ProductID

- ProductName
- Category (Furniture, Technology, Office Supplies)
- SubCategory

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### 5.4 Regions Table

- RegionID
- RegionName (East, West, North, South)
- City

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### 5.5 Date Table (Will be created in Power BI later)

- Date
- Year
- Month
- Month Name
- Quarter
- Day

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## 6. Sample Dataset

### Sales Table (Sample Rows)

| OrderID | OrderDate | CustomerID | ProductID | RegionID | Quantity | UnitPrice | Discount | SalesAmount | Profit |
|---|---|---|---|---|---|---|---|---|---|
| 1001 | 2023-01-05 | C001 | P001 | R1 | 2 | 500 | 0.1 | 900 | 200 |
| 1002 | 2023-01-06 | C002 | P002 | R2 | 1 | 1200 | 0.05 | 1140 | 300 |
| 1003 | 2023-01-07 | C003 | P003 | R3 | 5 | 100 | 0 | 500 | 150 |
| 1004 | 2023-02-01 | C001 | P002 | R1 | 3 | 1200 | 0.2 | 2880 | 600 |
| 1005 | 2023-02-10 | C004 | P004 | R4 | 4 | 250 | 0.1 | 900 | 200 |

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### Customers Table

| CustomerID | CustomerName | Segment |
|---|---|---|

| | | |
|---|---|---|
| C001 | Ravi Kumar | Consumer |
| C002 | Anil Sharma | Corporate |
| C003 | Sita Devi | Home Office |
| C004 | John Paul | Consumer |

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### Products Table

| ProductID | ProductName | Category | SubCategory |
|---|---|---|---|
| P001 | Laptop | Technology | Computers |
| P002 | Printer | Technology | Accessories |
| P003 | Notebook | Office Supplies | Paper |
| P004 | Chair | Furniture | Seating |

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### Regions Table

| RegionID | RegionName | City |
|---|---|---|
| R1 | South | Chennai |
| R2 | North | Delhi |
| R3 | West | Mumbai |
| R4 | East | Kolkata |

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## 7. Key Learning Concepts in This Part

- Understanding business requirements
- Designing a data model (fact vs dimension tables)
- Creating structured datasets
- Preparing for Power BI import

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# II. Importing Data into Power BI and Understanding the Interface

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## 1. Goal of This Part

- Import your Excel dataset into Power BI
- Understand the Power BI interface
- Explore the three main views
- Verify your data is loaded correctly

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## 2. Open Power BI Desktop

1. Launch **Power BI Desktop**
2. On the home screen, click:
   **Home → Get Data → Excel**

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## 3. Import Your Excel File

1. Select your Excel file (created in Part 1)
2. Navigator window will appear
3. Select all sheets:
   - Sales
   - Customers
   - Products
   - Regions
4. Click **Load**

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## 4. What Happens After Loading

Power BI will:

- Import data into its internal model
- Automatically detect data types
- Attempt basic relationships (may or may not be correct)

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# 5. Understanding the Power BI Interface

Power BI has **3 main views** (very important for beginners):

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## 5.1 Report View (Default)

This is where you build dashboards.

What you see:

- Blank canvas (report page)
- Fields pane (right side)
- Visualizations pane

Purpose:

- Create charts
- Build dashboard
- Add filters and slicers

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## 5.2 Data View

Click the **table icon** on the left sidebar.

What you see:

- Tabular format of your data

Purpose:

- Inspect rows and columns
- Validate data correctness
- Create calculated columns

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## 5.3 Model View

Click the **relationship icon**.

What you see:

- Diagram of tables and relationships

Purpose:

- Define relationships
- Build star schema
- Control data model behavior

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# 6. Verify Data After Import

Go to **Data View** and check:

### Sales Table

- Dates are in date format
- Numbers (SalesAmount, Profit) are numeric

### Customers Table

- CustomerID is text
- No missing values

### Products Table

- Categories and Subcategories are filled

### Regions Table

- Region names and cities are correct

---

# 7. Fix Common Data Issues

If something is wrong:

### Change Data Type

1. Select column
2. Go to **Column Tools**

3. Change Data Type:
- Date → Date
- Numbers → Decimal Number / Whole Number
- Text → Text

---

## 8. Rename Tables (Best Practice)

Rename tables for clarity:

- Sales → FactSales
- Customers → DimCustomers
- Products → DimProducts
- Regions → DimRegions

Steps:

1. Go to Fields pane
2. Right-click table
3. Rename

---

## 9. Hide Unnecessary Columns (Optional but Professional)

Example:

- Hide IDs not needed in visuals

Steps:

1. Right-click column
2. Click **Hide in Report View**

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## 10. Save Your Project

Save your file:

**File → Save As → Retail_Sales_Dashboard.pbix**

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## 11. Key Concepts Learned

- Data import process
- Power BI interface navigation
- Data validation
- Data types importance
- Basic data preparation

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## 12. Assignment

1. Import all tables successfully
2. Verify data types
3. Rename tables properly
4. Save your PBIX file

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# III. Data Modeling (Creating Relationships and Star Schema)

A correct data model ensures your reports work accurately and efficiently.

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## 1. Goal of This Part

- Understand relationships between tables
- Build a **star schema**
- Create correct relationships manually
- Avoid common beginner mistakes

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## 2. What is a Data Model?

A data model defines how tables are connected.

In Power BI:

- Tables do not automatically “know” each other
- Relationships allow filtering and aggregation across tables

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# 3. Understanding Star Schema (Core Concept)

You already created:

### Fact Table

- **FactSales** → contains transactions (numbers)

### Dimension Tables

- **DimCustomers**
- **DimProducts**
- **DimRegions**

In a star schema:

- Fact table is in the center
- Dimension tables surround it
- Relationships flow from dimensions → fact

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# 4. Go to Model View

1. Click **Model View (relationship icon)**
2. You will see all tables as boxes

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# 5. Create Relationships Manually

Power BI may create some automatically, but you must verify or rebuild them.

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### 5.1 Customer Relationship

Connect:

- DimCustomers[CustomerID] → FactSales[CustomerID]

Steps:

1. Drag **CustomerID** from DimCustomers
2. Drop onto **CustomerID** in FactSales

Settings:

- Cardinality: One-to-Many (1 → *)
- Cross filter: Single
- Active: Yes

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### 5.2 Product Relationship

Connect:

- DimProducts[ProductID] → FactSales[ProductID]

---

### 5.3 Region Relationship

Connect:

- DimRegions[RegionID] → FactSales[RegionID]

---

## 6. Relationship Rules (Very Important)

| Rule | Explanation |
|---|---|
| One-to-Many | Dimension → Fact |
| Single Direction | Avoid confusion for beginners |
| Unique Key | Dimension must have unique IDs |
| No loops | Avoid circular relationships |

---

## 7. Validate Your Model

Your model should look like:

- FactSales in center
- All dimensions connected to it
- No connections between dimension tables

---

# 8. Test Relationships (Simple Check)

Go to **Report View**:

1. Add a table visual
2. Drag:
    - CustomerName (DimCustomers)
    - SalesAmount (FactSales)

If relationships are correct:

- Sales values will appear correctly per customer

If wrong:

- Blank values or incorrect totals

---

# 9. Common Beginner Mistakes

## Mistake 1: Many-to-Many Relationships

Avoid unless absolutely necessary

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## Mistake 2: Connecting Dimension to Dimension

Wrong:

- DimCustomers → DimProducts

Correct:

- Only connect to FactSales

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### Mistake 3: Duplicate IDs in Dimension Tables

Ensure:

- CustomerID is unique in DimCustomers

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### Mistake 4: Wrong Data Types

CustomerID must be same type in both tables

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## 10. Why Star Schema Matters

Benefits:

- Faster performance
- Easier DAX calculations
- Accurate filtering
- Cleaner reports

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## 11. Assignment

1. Create all 3 relationships:
    - Customers → Sales
    - Products → Sales
    - Regions → Sales
2. Ensure:
    - One-to-Many relationships
    - Single direction filtering
3. Test with a simple table visual

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# IV. Creating a Calendar Table (Date Table) and Time Intelligence Foundation

Time-based analysis is one of the most powerful features in Power BI. Without a proper Date table, your reports will be limited and often incorrect.

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# 1. Goal of This Part

- Create a proper Calendar (Date) table using DAX
- Add useful time-related columns
- Connect it to your data model
- Prepare for time intelligence calculations

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# 2. Why Do We Need a Date Table?

Power BI requires a dedicated Date table to:

- Analyze trends (monthly, yearly)
- Compare periods (YoY, MoM)
- Enable time intelligence functions

Important rule:

There must be **one continuous date table** connected to your fact table.

---

# 3. Create Calendar Table Using DAX

Go to:

**Modeling → New Table**

Enter the following DAX:

```
DimDate =
ADDCOLUMNS (
  CALENDAR (DATE(2023,1,1), DATE(2024,12,31)),
  "Year", YEAR([Date]),
  "Month Number", MONTH([Date]),
  "Month Name", FORMAT([Date], "MMMM"),
  "Year-Month", FORMAT([Date], "YYYY-MM"),
  "Quarter", "Q" & FORMAT([Date], "Q"),
```

```
  "Day", DAY([Date]),
  "Day Name", FORMAT([Date], "dddd")
)
```

---

## 4. Explanation of DAX

- **CALENDAR()** → Creates continuous date range
- **ADDCOLUMNS()** → Adds extra columns
- **YEAR(), MONTH(), DAY()** → Extract date parts
- **FORMAT()** → Converts to readable labels

---

## 5. Mark as Date Table (Critical Step)

1. Select **DimDate table**
2. Go to **Table Tools**
3. Click **Mark as Date Table**
4. Select column: **Date**

This enables time intelligence functions.

---

## 6. Create Relationship with Fact Table

Go to **Model View**:

Connect:

- DimDate[Date] → FactSales[OrderDate]

Settings:

- One-to-Many (1 → *)
- Single direction
- Active

---

## 7. Sort Month Name Properly

Problem:

- Months appear alphabetically (April, August…)

Solution:

1. Select **Month Name**
2. Click **Sort by Column**
3. Choose **Month Number**

---

## 8. Hide Unnecessary Columns

You may hide:

- Month Number
- Date (optional depending on use)

---

## 9. Test Date Table

Go to Report View:

Create a table:

- Year
- Month Name
- SalesAmount

If working correctly:

- Data should aggregate properly by month and year

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## 10. Key Concepts Learned

- Importance of Date table
- Creating calculated tables in DAX
- Time hierarchy (Year → Month → Day)
- Sorting columns correctly

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## 11. Assignment

1. Create DimDate using DAX
2. Mark it as Date Table
3. Create relationship with FactSales
4. Sort Month Name correctly
5. Test with a visual

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# V. Creating Basic Measures Using DAX

Now you move into the most important skill in Power BI: **DAX (Data Analysis Expressions)**. Measures are the foundation of all dashboards.

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## 1. Goal of This Part

- Understand **measures vs calculated columns**
- Create core business metrics
- Use basic DAX aggregation functions
- Build reusable calculations

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## 2. Measures vs Calculated Columns

### Calculated Column

- Row-level calculation
- Stored in the table
- Example: Profit per row

### Measure

- Aggregated calculation
- Computed dynamically based on filters
- Used in visuals

Important rule:

**Dashboards use measures, not columns**

---

## 3. Where to Create Measures

Best practice:

- Create a separate table called **Measures**

### Steps:

1. Go to **Modeling** → **New Table**
2. Enter:

Measures = {}

3. Select this table before creating measures

---

## 4. Create Your First Measure — Total Sales

Go to **Modeling** → **New Measure**

Total Sales = SUM(FactSales[SalesAmount])

---

## 5. Create Total Quantity

Total Quantity = SUM(FactSales[Quantity])

---

## 6. Create Total Profit

Total Profit = SUM(FactSales[Profit])

---

## 7. Create Count of Orders

Total Orders = COUNT(FactSales[OrderID])

---

## 8. Create Average Sales per Order

Avg Sales per Order = DIVIDE([Total Sales], [Total Orders])

Why use `DIVIDE()` instead of `/`?

- Handles division by zero safely

---

## 9. Format Measures

For each measure:

1. Select measure
2. Go to **Measure Tools**

Set:

- Currency → Total Sales, Total Profit
- Whole Number → Total Quantity
- Decimal → Avg Sales

---

## 10. Test Measures in Visuals

Go to **Report View**

Create a **Table Visual**:

Add:

- CustomerName
- Total Sales
- Total Profit

Expected result:

- Values should change dynamically when filtered

---

## 11. Understand Filter Context (Critical Concept)

Measures change based on:

- Filters
- Slicers
- Visuals

Example:

If you filter Region = South
→ Total Sales shows only South data

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# 12. Common Beginner Mistakes

### Mistake 1: Using Columns Instead of Measures

Wrong:

- Drag SalesAmount directly

Correct:

- Use [Total Sales]

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### Mistake 2: Creating Measures in Fact Table

Better:

- Keep measures in a separate table

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### Mistake 3: Using COUNT instead of COUNTROWS

Better version:

Total Orders = COUNTROWS(FactSales)

---

# 13. Assignment

1. Create all measures:

    - Total Sales
    - Total Quantity
    - Total Profit
    - Total Orders
    - Avg Sales per Order
2. Format them properly
3. Test in:
    - Table visual
    - Card visual

---

# VI. Creating Calculated Columns and Derived Metrics

In this part, you will learn how to create **row-level calculations** using DAX. These are called **calculated columns** and are different from measures.

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## 1. Goal of This Part

- Understand when to use calculated columns
- Create derived fields like Profit Margin
- Analyze discounts and pricing
- Prepare data for better visuals

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## 2. When to Use Calculated Columns

Use calculated columns when:

- You need **row-by-row computation**
- You want to create **categories or flags**
- You need columns for slicing or grouping

Do NOT use them for aggregations (that is for measures)

---

## 3. Where to Create Calculated Columns

Go to:

- **Data View**
- Select the table (mostly **FactSales**)
- Click **New Column**

---

## 4. Create Profit Margin Column

Profit Margin = DIVIDE(FactSales[Profit], FactSales[SalesAmount])

Explanation:

- Calculates profit % per row
- Uses DIVIDE for safety

---

## 5. Create Discount Amount

Discount Amount = FactSales[UnitPrice] * FactSales[Quantity] * FactSales[Discount]

---

## 6. Create Net Sales

Net Sales = FactSales[UnitPrice] * FactSales[Quantity] - FactSales[Discount Amount]

---

## 7. Create Price Category

```
Price Category =
IF(
  FactSales[UnitPrice] > 500,
  "High Value",
  "Low Value"
)
```

---

## 8. Create Order Size Category

Order Size =

```
SWITCH(
  TRUE(),
  FactSales[Quantity] >= 5, "Large",
  FactSales[Quantity] >= 2, "Medium",
  "Small"
)
```

---

## 9. Format Columns

- Profit Margin → Percentage
- Discount Amount → Currency
- Net Sales → Currency

---

## 10. Use Columns in Visuals

Go to **Report View**:

Create visuals like:

### Table Visual

- ProductName
- Price Category
- Net Sales

### Bar Chart

- Order Size
- Total Sales

---

## 11. Key Difference Reinforced

| Feature | Calculated Column | Measure |
|---|---|---|
| Level | Row | Aggregated |
| Storage | Stored in model | Calculated on demand |

| Use case | Categories | KPIs |
|---|---|---|

---

# 12. Common Beginner Mistakes

### Mistake 1: Using Measures Inside Columns Incorrectly

Avoid mixing contexts without understanding

---

### Mistake 2: Creating Too Many Columns

Leads to:

- Large file size
- Slow performance

---

### Mistake 3: Repeating Existing Calculations

Example:

- SalesAmount already exists → don’t recreate unnecessarily

---

# 13. Assignment

1. Create all calculated columns:
    - Profit Margin
    - Discount Amount
    - Net Sales
    - Price Category
    - Order Size
2. Format them properly
3. Test in at least:
    - One table visual
    - One chart

---

# VII. Building Your First Dashboard (Visuals and Layout)

This is where everything starts coming together. You will now create your first complete Power BI report page using the measures and columns you built.

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## 1. Goal of This Part

- Create key visuals (cards, charts, tables)
- Design a clean dashboard layout
- Use slicers for interactivity
- Build your first complete report page

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## 2. Plan Your Dashboard Layout

Before building visuals, think like a business analyst.

### Recommended Layout:

Top Section (KPIs):

- Total Sales
- Total Profit
- Total Orders

Middle Section:

- Sales by Category
- Sales by Region

Bottom Section:

- Sales Trend (over time)
- Detailed table

---

## 3. Create KPI Cards

Go to **Report View**

### Add Card Visuals:

**Card 1 — Total Sales**

- Visual: Card
- Field: [Total Sales]

**Card 2 — Total Profit**

- Field: [Total Profit]

**Card 3 — Total Orders**

- Field: [Total Orders]

Format:

- Increase font size
- Add titles

---

## 4. Create Sales by Category (Bar Chart)

### Visual: Clustered Bar Chart

Fields:

- Axis → DimProducts[Category]
- Values → [Total Sales]

Purpose:

- Compare categories easily

---

## 5. Create Sales by Region (Column Chart)

### Visual: Clustered Column Chart

Fields:

- Axis → DimRegions[RegionName]
- Values → [Total Sales]

---

# 6. Create Sales Trend (Line Chart)

### Visual: Line Chart

Fields:

- Axis → DimDate[Year-Month]
- Values → [Total Sales]

Important:

- Ensure Month is sorted correctly (done in Part 4)

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# 7. Create Top Products Table

### Visual: Table

Fields:

- ProductName
- [Total Sales]
- [Total Profit]

Sort:

- By Total Sales (Descending)

---

# 8. Add Slicers (Filters for Users)

### Slicer 1 — Region

- Field → RegionName

### Slicer 2 — Category

- Field → Category

### Slicer 3 — Year

- Field → Year

Purpose:

- Allow user to filter entire dashboard

---

## 9. Arrange the Layout

Best practices:

- Align visuals properly
- Keep equal spacing
- Use grid layout
- Avoid clutter

Suggested structure:

Row 1 → KPI Cards
Row 2 → Bar + Column charts
Row 3 → Line chart
Row 4 → Table

---

## 10. Basic Formatting

Select each visual → Format pane:

Apply:

- Titles (clear and meaningful)
- Data labels (optional)
- Background (light color)
- Borders (optional)

---

## 11. Interaction Check

Test your dashboard:

1. Click a region in chart
2. See if other visuals update

If yes → interactions working correctly

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## 12. Rename Report Page

At bottom:

- Rename to: **Sales Overview Dashboard**

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## 13. Key Concepts Learned

- Visual selection based on purpose
- Layout design
- Interactivity using slicers
- Dashboard storytelling basics

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## 14. Assignment

1. Create all visuals:
    - 3 KPI cards
    - 2 charts
    - 1 line chart
    - 1 table
2. Add at least 2 slicers
3. Arrange layout neatly

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# VIII. Advanced Visuals and Filtering Techniques

Now that your basic dashboard is ready, you will enhance it with **advanced features** that make your report interactive, insightful, and closer to real-world business dashboards.

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# 1. Goal of This Part

- Apply **Top N filtering**
- Use **drill-down functionality**
- Add **tooltips** for better insights
- Use **conditional formatting**
- Improve user experience

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# 2. Top N Filtering (Very Important for Business)

### Example: Top 10 Products by Sales

**Steps:**

1. Select your **Table or Bar Chart**
2. Drag **ProductName** into Filters pane
3. Change filter type to **Top N**
4. Set:
    - Top → 10
    - By Value → [Total Sales]
5. Click **Apply Filter**

Result:

- Only top-performing products are shown

---

# 3. Drill-Down Functionality

Allows users to explore deeper levels of data.

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### Example: Category → SubCategory

**Steps:**

1. Select Bar Chart

2. Add fields:
- Axis:
    - Category
    - SubCategory
3. Enable drill-down (top-left icon on visual)

Now user can:

- Click category → drill into subcategories

---

# 4. Drill-Up and Drill Mode

- Drill Down → Go deeper
- Drill Up → Go back
- Expand → Show all levels together

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# 5. Add Tooltips (Hidden Insight Feature)

Tooltips show extra information when hovering.

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### Create Tooltip Page

1. Add new page
2. Rename: **Tooltip Page**
3. Go to Format:
    - Page Size → Tooltip
    - Turn ON Tooltip

---

### Add Visuals to Tooltip

Example:

- Add:
    - Total Sales
    - Total Profit
    - Category

---

### Assign Tooltip to Visual

1. Select main chart
2. Format → Tooltip
3. Choose **Tooltip Page**

Result:

- Hover over chart → see detailed info

---

## 6. Conditional Formatting

Highlights important data visually.

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### Example: Color Based on Sales

**Steps:**

1. Select Table
2. Click dropdown on **Total Sales**
3. Conditional Formatting → Background Color

Set rules:

- High sales → Green
- Low sales → Red

---

### Example: Data Bars

- Add bars inside table cells to visualize magnitude

---

## 7. Use Filters Pane Effectively

Types of filters:

### Visual Level Filter

- Applies to one visual

### Page Level Filter

- Applies to entire page

### Report Level Filter

- Applies to all pages

---

## 8. Add a Top Customers Visual

### Create Bar Chart:

Fields:

- CustomerName
- [Total Sales]

Apply:

- Top N → Top 5 customers

---

## 9. Enable Sorting

Example:

- Sort charts by:
    - Total Sales (Descending)

Steps:

1. Click visual
2. Click 3 dots
3. Sort by measure

---

# 10. Improve User Experience

Add:

- Clear titles
- Consistent formatting
- Logical grouping
- Avoid overcrowding

---

# 11. Common Beginner Mistakes

### Mistake 1: Too Many Visuals

Leads to confusion

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### Mistake 2: No Filtering

Dashboard becomes static

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### Mistake 3: Ignoring Drill Features

Miss deeper insights

---

# 12. Assignment

1. Apply Top N filter:
    - Top 10 Products
    - Top 5 Customers
2. Add drill-down:
    - Category → SubCategory
3. Create one tooltip page
4. Apply conditional formatting

---

# IX. Time Intelligence (Year-over-Year, Month-over-Month Analysis)

This is where your dashboard becomes truly analytical. You will now use DAX to compare performance across time.

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## 1. Goal of This Part

- Create **Year-to-Date (YTD)** calculations
- Compare **current vs previous year**
- Calculate **growth percentages**
- Analyze trends over time

---

## 2. Important Requirement

Time intelligence works ONLY if:

- You have a proper **DimDate table**
- It is **marked as Date Table**
- It is **connected to FactSales[OrderDate]**

---

## 3. Create Year-to-Date (YTD) Sales

Go to **Measures table → New Measure**

$YTD\ Sales = TOTALYTD([Total\ Sales], DimDate[Date])$YTD Sales=TOTALYTD([Total Sales], DimDate[Date])

Explanation:

- Calculates cumulative sales from start of year to selected date

---

## 4. Create Previous Year Sales

$Previous\ Year\ Sales$=$CALCULATE$([$Total\ Sales$], $SAMEPERIODLASTYEAR$($DimDate$[$Date$]))Previous Year Sales=CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DimDate[Date]))

---

# 5. Create Year-over-Year Growth

$YoY\ Growth$ %=$DIVIDE$([$Total\ Sales$]−[$Previous\ Year\ Sales$], [$Previous\ Year\ Sales$])YoY Growth %=DIVIDE([Total Sales]−[Previous Year Sales], [Previous Year Sales])

Format:

- Percentage

---

# 6. Create Month-over-Month (MoM) Growth

$Previous\ Month\ Sales$=$CALCULATE$([$Total\ Sales$], $DATEADD$($DimDate$[$Date$], −1, $MONTH$))Previous Month Sales=CALCULATE([Total Sales], DATEADD(DimDate[Date], −1, MONTH))

$MoM\ Growth$ %=$DIVIDE$([$Total\ Sales$]−[$Previous\ Month\ Sales$], [$Previous\ Month\ Sales$])MoM Growth %=DIVIDE([Total Sales]−[Previous Month Sales], [Previous Month Sales])

---

# 7. Add Time Intelligence to Visuals

## Line Chart (Trend Analysis)

Fields:

- Axis → DimDate[Year-Month]
- Values:
    - Total Sales
    - Previous Year Sales

---

## KPI Cards

Add cards for:

- YTD Sales
- YoY Growth %

---

# 8. Create Comparison Table

### Table Visual:

Fields:

- Year
- Month Name
- Total Sales
- Previous Year Sales
- YoY Growth %

---

# 9. Business Insights You Can Now Answer

- Are sales growing year over year?
- Which months perform best?
- Is growth consistent or seasonal?
- Are we improving compared to last year?

---

# 10. Common Beginner Mistakes

### Mistake 1: Missing Date Table

Time intelligence will not work

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### Mistake 2: Wrong Relationship

Date must connect to FactSales[OrderDate]

---

### Mistake 3: Using Text Dates

Date must be true date type

---

### Mistake 4: Not Marking Date Table

Required for functions like TOTALYTD

---

## 11. Assignment

1. Create measures:
    - YTD Sales
    - Previous Year Sales
    - YoY Growth %
    - MoM Growth %
2. Add to:
    - Line chart
    - Table
    - KPI cards
3. Format percentages properly

---

# X. Dashboard Design, Themes, and User Experience

Now you will transform your dashboard from “working” to **professional and presentation-ready**. This is what differentiates beginners from real analysts.

---

## 1. Goal of This Part

- Apply consistent design principles
- Use themes and colors effectively
- Improve readability and usability
- Make your dashboard business-friendly

---

# 2. Principles of Good Dashboard Design

Follow these core principles:

### Clarity

- Easy to understand at a glance

### Consistency

- Same colors, fonts, and layout

### Focus

- Highlight important metrics

### Simplicity

- Avoid unnecessary visuals

---

# 3. Apply a Theme

Power BI provides built-in themes.

### Steps:

1. Go to **View** → **Themes**
2. Choose a theme (e.g., Default, Dark, Colorful)

---

### Custom Theme (Optional)

You can define:

- Primary color (e.g., blue)
- Accent color (e.g., orange)
- Background color (light grey or white)

---

# 4. Color Best Practices

Use colors with meaning:

- Green → Positive (profit, growth)
- Red → Negative (loss, decline)
- Blue → Neutral metrics

Avoid:

- Too many colors
- Random color usage

---

# 5. Layout Improvements

## Align and Distribute

1. Select multiple visuals
2. Use:
    - Align (Top, Left, Center)
    - Distribute (Horizontal/Vertical)

---

## Use Grid Structure

Divide your report into sections:

- Top → KPIs
- Middle → Comparisons
- Bottom → Details

---

# 6. Add Titles and Labels

Each visual should answer a question.

Bad:

- “Sales”

Good:

- “Total Sales by Category”

---

## 7. Improve KPI Cards

Enhance cards:

- Increase font size
- Add background color
- Add labels (e.g., “Total Sales (₹)”)

---

## 8. Use Icons and Indicators (Optional)

Example:

- Up arrow → growth
- Down arrow → decline

Can be done using:

- Conditional formatting
- Unicode symbols

---

## 9. Consistent Formatting Across Visuals

Ensure:

- Same font style
- Same number format
- Same color logic

Example:

- Sales always blue
- Profit always green

---

## 10. Add Page Navigation (Optional Advanced Step)

If multiple pages:

1. Insert → Buttons
2. Add navigation buttons:
    - Overview
    - Product Analysis
    - Region Analysis

---

## 11. Remove Clutter

Remove:

- Unnecessary gridlines
- Extra labels
- Duplicate visuals

Keep only what adds value

---

## 12. Add Report Title and Branding

At top of page:

- “Retail Sales Performance Dashboard”

Optional:

- Add company name/logo

---

## 13. Test User Experience

Ask:

- Can a manager understand this in 10 seconds?
- Are key insights obvious?
- Are filters easy to use?

---

## 14. Common Beginner Mistakes

### Mistake 1: Overdesign

Too many colors and visuals

---

### Mistake 2: No Alignment

Dashboard looks unprofessional

---

### Mistake 3: Poor Titles

Users don’t understand visuals

---

## 15. Assignment

1. Apply a theme
2. Align all visuals properly
3. Add meaningful titles
4. Standardize colors
5. Improve KPI card design

---

# XI. Publishing, Sharing, and Refreshing Data

Now your dashboard is ready. In this part, you will learn how to move from **development (Power BI Desktop)** to **real-world usage (Power BI Service)**.

---

## 1. Goal of This Part

- Publish your report to Power BI Service

- Share dashboards with others
- Understand workspaces
- Configure data refresh

---

## 2. What is Power BI Service?

Power BI Service is the **cloud platform** where:

- Reports are hosted
- Dashboards are shared
- Data is refreshed
- Teams collaborate

You access it via browser:

- https://app.powerbi.com

---

## 3. Publish Your Report

### Steps:

1. Open your `.pbix` file
2. Click **Home** → **Publish**
3. Sign in (if not already)
4. Select destination:
    - My Workspace (for beginners)
5. Click **Select**

---

## 4. Verify in Power BI Service

1. Go to browser → Power BI Service
2. Open **My Workspace**
3. You will see:

- Dataset
- Report

Click report → verify visuals

---

# 5. Create a Dashboard (Pin Visuals)

In Power BI Service:

1. Open your report
2. Hover over a visual
3. Click **Pin icon**
4. Choose:

- New Dashboard → Name it: *Retail Dashboard*

---

# 6. Share the Dashboard

## Option 1: Share Directly

1. Click **Share**
2. Enter email IDs
3. Set permissions:
    - View only
    - Allow sharing (optional)

---

## Option 2: Workspace Sharing (Better for Teams)

- Add users to workspace
- Control access centrally

---

# 7. Understand Dataset and Report

- Dataset → Data model + measures
- Report → Visual layer

Important:

- Multiple reports can use one dataset

---

# 8. Data Refresh

### Why Refresh?

Your Excel data may change:

- New sales
- Updated records

---

### Types of Refresh:

1. Manual Refresh
2. Scheduled Refresh

---

# 9. Configure Scheduled Refresh

### Steps:

1. Go to **Dataset settings**
2. Click **Scheduled Refresh**
3. Turn ON
4. Set frequency:
- Daily (recommended for beginners)

---

# 10. Gateway (Important Concept)

If your data is:

- Local file (Excel on your computer)

You need:

- **On-Premises Data Gateway**

For beginners:

- Keep file in **OneDrive** or **SharePoint**
  → No gateway required

---

## 11. Export and Sharing Options

You can:

- Export to PDF
- Export to PowerPoint
- Share link

---

## 12. Security Basics

Control access:

- Who can view
- Who can edit
- Who can share

---

## 13. Common Beginner Mistakes

### Mistake 1: Not Publishing After Changes

Always republish after updates

---

### Mistake 2: Broken Refresh

Occurs when file path changes

---

### Mistake 3: Sharing Without Permissions

Users cannot access data

---

## 14. Assignment

1. Publish your report
2. Open it in Power BI Service
3. Create a dashboard (pin visuals)
4. Share with at least one user (or test account)
5. Set up scheduled refresh

---

# XII. Final Project Enhancement and Real-World Best Practices

This is the final stage where you refine your project to a **professional, production-ready Power BI solution**. You will focus on performance, maintainability, and real-world standards.

---

## 1. Goal of This Part

- Optimize your data model
- Improve performance
- Apply best practices used in real companies
- Validate your dashboard as a complete project

---

## 2. Final Data Model Review

Re-check your model:

### Must-have structure:

- FactSales (center)
- DimCustomers
- DimProducts
- DimRegions
- DimDate

Checklist:

- All relationships are **1 → ***
- No many-to-many relationships
- No circular relationships

- Proper data types

---

# 3. Optimize Data Model

## Remove Unused Columns

- Delete or hide columns not used in visuals or calculations

---

## Reduce Data Size

- Avoid unnecessary calculated columns
- Keep only required fields

---

## Use Measures Instead of Columns

Better:

- Measures (dynamic, efficient)

Avoid:

- Too many calculated columns

---

# 4. Improve DAX Performance

## Use Simple Measures

Good:

Total Sales = SUM(FactSales[SalesAmount])

Avoid:

- Complex nested formulas unless needed

---

### Use Variables (Advanced but Recommended)

```
Profit % =
VAR ProfitValue = [Total Profit]
VAR SalesValue = [Total Sales]
RETURN
DIVIDE(ProfitValue, SalesValue)
```

Benefits:

- Cleaner code
- Faster performance
- Easier debugging

---

## 5. Naming Conventions (Professional Standard)

Use clear names:

### Tables:

- FactSales
- DimCustomers

### Measures:

- Total Sales
- Total Profit
- YoY Growth %

Avoid:

- Measure1, Column1

---

## 6. Organize Fields

### Create Folders for Measures

Steps:

1. Select a measure
2. In Properties → Display Folder

3. Group like:
- Sales Metrics
- Time Intelligence
- KPIs

---

# 7. Add Final Dashboard Enhancements

### KPI Section Improvements

- Add:
    - YTD Sales
    - YoY Growth %
    - MoM Growth %

---

### Add Insight Text (Optional but Powerful)

Example:

- “Sales increased by 12% compared to last year”

---

# 8. Validate Business Questions

Your dashboard must answer:

- What is total sales and profit?
- Which products perform best?
- Which regions are strongest?
- Are sales increasing over time?
- Who are top customers?

If any answer is missing → improve visuals

---

# 9. Performance Testing

Check:

- Does report load quickly?
- Do visuals respond instantly?
- Are filters smooth?

If slow:

- Reduce visuals
- Simplify DAX
- Optimize model

---

## 10. Documentation (Very Important in Real Jobs)

Prepare a simple document:

### Include:

- Data sources
- Table descriptions
- Measure definitions
- Business logic

---

## 11. Final Project Checklist

Before considering your project complete:

### Data

- Clean and structured

### Model

- Star schema

### DAX

- Core measures created

### Visuals

- Clear and meaningful

### Design

- Professional layout

### Interactivity

- Slicers and filters working

### Time Intelligence

- YTD, YoY implemented

### Deployment

- Published to Power BI Service

---

## 12. Real-World Best Practices

- Always start with business questions
- Keep model simple
- Avoid overloading dashboard
- Use consistent naming
- Test with real users

---

## 13. Next Steps for Learning

After this project, you can move to:

- Advanced DAX (CALCULATE, FILTER)
- Power Query (data transformation)
- Row-Level Security (RLS)
- Large dataset optimization
- Real-time dashboards

---

## 14. Final Assignment

Complete your project by:

1. Reviewing and optimizing your model
2. Cleaning unused fields
3. Improving dashboard design
4. Adding advanced KPIs
5. Publishing final version

---

# 15. Project Completion

You have now built a **complete end-to-end Power BI project**:

- Data → Model → DAX → Dashboard → Deployment

This is the exact workflow used in real companies.